Meihao Liao

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Hong Kong Baptist University, Hong Kong

Education

- **Beijing Institute of Technology** *September 2020 - June 2026*
  Ph.D. in Computer Science Beijing, China
  - Supervisor: Prof. Rong-Hua Li
- **Beijing Institute of Technology** *September 2016 - June 2020*
  B.Eng. in Computer Science and Engineering Beijing, China

Research Experience

- **Hong Kong Baptist University** *September 2026 - Present*
  Postdoctoral Researcher Hong Kong
  - Supervisor: Prof. Xin Huang

Research Direction

My general research interest lies in spectral graph theory, numerical linear algebra and their applications in graph data management, graph machine learning, etc. I'm interested in efficient implementations of theoretically fast algorithms.

Publications

C=Conference, J=Journal, S=In Submission, T=Thesis

[C.1] **Meihao Liao**, Bao Xing, Rong-Hua Li, Guoren Wang. Tree+DAG: An Efficient Index for Resistance Distance Computation on Core-Periphery Graphs. In SIGMOD, 2027.

[C.2] **Meihao Liao**, Yueyang Pan, Rong-Hua Li, Guoren Wang. Efficient Exact Resistance Distance Computation on Small-Treewidth Graphs: A Labelling Approach. In SIGMOD, 2026.

[C.3] **Meihao Liao**, Cheng Li, Rong-Hua Li, Guoren Wang. Efficient Index Maintenance for Effective Resistance Computation on Evolving Graphs. In SIGMOD, 2025.

[C.4] **Meihao Liao**, Junjie Zhou, Rong-Hua Li, Qiangqiang Dai, Hongyang Chen, Guoren Wang. Efficient and Provable Effective Resistance Computation on Large Graphs: an Index-based Approach. In SIGMOD, 2024.

[C.5] **Meihao Liao**, Rong-Hua Li, Qiangqiang Dai, Hongyang Chen, Hongchao Qin, Guoren Wang. Efficient Resistance Distance Computation: The Power of Landmark-based Approaches. In SIGMOD, 2023.

[C.6] **Meihao Liao**, Rong-Hua Li, Qiangqiang Dai, Hongyang Chen, Hongchao Qin, Guoren Wang. Efficient Personalized PageRank Computation: The Power of Variance-Reduced Monte Carlo Approaches. In SIGMOD, 2023.

[C.7] **Meihao Liao**, Rong-Hua Li, Qiangqiang Dai, Guoren Wang. Efficient Personalized PageRank Computation: A Spanning Forests Sampling Based Approach. In SIGMOD, 2022.

[J.1] **Meihao Liao**, Rong-Hua Li, Qiangqiang Dai, Guoren Wang. Scaling up Electrical Closeness Centrality Computation. In TKDE, 2026.

- **[C.8]** Junjie Zhou, **Meihao Liao**, Rong-Hua Li, Longlong Lin, Guoren Wang. One Index for All: Towards Efficient Personalized PageRank Computation for Every Damping Factor. In SIGMOD, 2026.
- **[C.9]** Yichun Yang, Longlong Lin, Rong-Hua Li, **Meihao Liao**, Guoren Wang. Theoretically and Practically Efficient Resistance Distance Computation on Large Graphs. In VLDB, 2026.
- **[C.10]** Zening Li, Rong-Hua Li, Yichun Yang, **Meihao Liao**. ID2GL: Incremental Node Representation Learning on Large-Scale Dynamic Directed Graphs. In CIKM, 2026. (Accepted.)
- **[C.11]** Yichun Yang, Rong-Hua Li, **Meihao Liao**, Guoren Wang. Improved Algorithms for Effective Resistance Computation on Graphs. In COLT, 2025.
- **[C.12]** Xunkai Li, **Meihao Liao**, Zhengyu Wu, Daohan Su, Wentao Zhang, Rong-Hua Li, Guoren Wang. LightDiC: A Simple yet Effective Approach for Large-scale Digraph Representation Learning. In VLDB, 2024.
- **[C.13]** Qiangqiang Dai, Rong-Hua Li, Donghang Cui, **Meihao Liao**, Yu-Xuan Qiu, Guoren Wang. Efficient Maximal Biplex Enumerations with Improved Worst-Case Time Guarantee. In SIGMOD, 2024.
- **[C.14]** Zening Li, Rong-Hua Li, **Meihao Liao**, Fusheng Jin, Guoren Wang. Privacy-Preserving Graph Embedding based on Local Differential Privacy. In CIKM, 2024.
- **[C.15]** Qiangqiang Dai, Rong-Hua Li, **Meihao Liao**, Guoren Wang. Maximal Defective Clique Enumeration. In SIGMOD, 2023.
- **[C.16]** Qiangqiang Dai, Rong-Hua Li, Xiaowei Ye, **Meihao Liao**, Weipeng Zhang, Guoren Wang. Hereditary Cohesive Subgraphs Enumeration on Bipartite Graphs: The Power of Pivot-based Approaches. In SIGMOD, 2023.
- **[C.17]** Qiangqiang Dai, Rong-Hua Li, **Meihao Liao**, Hongzhi Chen, Guoren Wang. Fast Maximal Clique Enumeration on Uncertain Graphs: A Pivot-based Approach. In SIGMOD, 2022.
- **[C.18]** Qiangqiang Dai, Rong-Hua Li, Hongchao Qin, **Meihao Liao**, Guoren Wang. Scaling Up Maximal k-plex Enumeration. In CIKM, 2022.
- **[J.2]** Yichun Yang, Rong-Hua Li, **Meihao Liao**, Guoren Wang. AdaPush: An Adaptive Push Framework for Graph-Propagation Based Node Similarity Computation. In TKDE, 2026.
- **[S.1]** Yueyang Pan, **Meihao Liao**, Rong-Hua Li. BD-Index: Scalable Biharmonic Distance Queries on Large Graphs via Divide-and-Conquer Indexing. In Submission.
- **[S.2]** Cheng Li, **Meihao Liao**, Rong-Hua Li, Guoren Wang. Scalable and Provable Kemeny Constant Computation on Static and Dynamic Graphs: A 2-Forest Sampling Approach. In Submission.
- **[S.3]** Yichun Yang, Rong-Hua Li, **Meihao Liao**, Guoren Wang. Scaling Up Graph Propagation Computation on Large Graphs: A Local Chebyshev Approximation Approach. In Submission.
- **[S.4]** Yichun Yang, Rong-Hua Li, **Meihao Liao**, Longlong Lin, Guoren Wang. Chebyshev Meets Push: Efficient and Provable Graph Propagation Computation. In Submission.
- **[S.5]** Rong-Hua Li, **Meihao Liao**, Yichun Yang. Classical and Quantum Spectral Density Estimation under Local Graph Access. In Submission.

Honors and Awards

- **National Scholarship (¥30,000)** *December, 2025*
  Ministry of Education of the People's Republic of China

- **National Scholarship (¥30,000)** *December, 2023*
  Ministry of Education of the People's Republic of China
- **Baidu Scholarship Nominee (20 people in the world)** *January, 2025*
  Baidu Company
- **ByteDance Scholarship Nominee (40 people in the world)** *December, 2024*
  ByteDance Company
- **Special Academic Scholarship** *2023 - 2026*
  Beijing Institute of Technology
- **Doctoral Dissertation Seed Fund (¥10,000)** *September, 2025*
  Beijing Institute of Technology

ACADEMIC SERVICE

- **Program Committee Member: KDD, WWW, ICML, NIPS, AAAI**
- **Journal Reviewer: TODS, TKDE, TKDD, Journal of Supercomputing**